

Culturally-Aware Image Captioning for Indigenous Languages of the Americas:

A Two-Stage Vision-and-Retrieval Pipeline

Romit Basak and Mehak Santosh Talmale and Nandita Mahendra and Tisha Patel

Khoury College of Computer Sciences, Northeastern University

{basak.r, talmale.m, mahendra.na, patel.tisha}@northeastern.edu

Abstract

Generic vision–language models (VLMs) discard culturally specific visual information before it can reach a downstream translator, a bottleneck exposed but left unresolved by the winning system of the AmericasNLP 2026 Cultural Image Captioning for Indigenous Languages (CICIL) shared task. We describe a two-stage pipeline that attacks this gap at the vision stage: Stage 1 uses a structured *cultural visual-question-answering* (VQA) prompt to elicit a culturally grounded Spanish description together with per-category cultural annotations; Stage 2 uses those annotations as retrieval keys and translates to the target language with Gemini 2.5 Flash. We report preliminary results over five languages (Guaraní, Bribri, Yucatec Maya, Wixárika, Nahuatl). On an intrinsic Spanish-side proxy (Wixárika pilot, $n=20$) concise cultural-VQA matches a generic baseline (20.9 vs. 21.0 ChrF++). End-to-end, cultural-VQA yields no consistent improvement over the generic control: four of five languages fall within ± 0.6 ChrF++ (noise level) and only Nahuatl shows a clear gain (+1.56), for a mean Δ of +0.20. We discuss why—including an incomplete retrieval bank and greedy-decoding degeneration on the lowest-resource languages—and outline the controlled experiments these preliminary findings motivate.

1 Introduction

The CICIL shared task (AmericasNLP 2026 Organizers, 2026) asks systems to caption culturally situated images directly in an indigenous language of the Americas (Guaraní, Yucatec Maya, Bribri, or Wixárika). The winning “Gators” submission (Dhawan et al., 2026) established a strong two-stage recipe—generate a Spanish intermediate description with a generic VLM, then translate with a retrieval-augmented LLM—but its own analysis identifies the *vision* stage as the bottleneck: a generic Qwen2.5-VL encoder (Qwen Team, 2025)

produces the Spanish intermediate, so culturally specific semantics are stripped out before translation begins.

We target exactly this gap. Our pipeline keeps the two-stage structure but makes Stage 1 culturally aware through a structured cultural-VQA prompt that interrogates the image (ceremony, material culture, landscape, kinship) before synthesizing a Spanish description, and emits the per-category answers as structured annotations that become Stage 2 retrieval keys. We ask three research questions. **RQ1:** does culturally-aware vision encoding improve end-to-end caption quality over a generic-VLM baseline? **RQ2:** does the benefit vary with language resource level? **RQ3:** which cultural categories are hardest to ground, and does this track caption quality? This preliminary paper contributes (i) the pipeline and its cultural-VQA module, (ii) an exploratory analysis of the CICIL data, and (iii) first end-to-end baseline numbers—reported honestly, including where the intervention does *not* help.

2 Related Work

2.1 Vision and Cultural Grounding

Generic VLMs trained on Western, English-dominated web data carry systematic cultural blind spots. Contrastive pretraining with CLIP (Radford et al., 2021) yields cross-modal representations skewed toward that distribution, and Liu et al. (2021) formalize the downstream cost through MaRVL, showing that cross-lingual, cross-cultural visual reasoning falls far short of English baselines. Romero et al. (2024) confirm with CVQA—co-created with native speakers across 30 countries—that culturally diverse visual QA remains open for state-of-the-art models. This representational bottleneck is what Dhawan et al. (2026) uncover but leave unresolved.

Prior interventions typically act at the vision

encoder. [Huang et al. \(2025\)](#) build CulTwin, a synthetic set of visually similar but culturally distinct triplets, and fine-tune CLIP with a tailored contrastive objective for up to 5.49% gains in fine-grained cultural concept recognition. Contrastive encoder retraining, however, needs large curated image sets that our ~ 50 -example-per-language budget cannot support; we therefore freeze the SigLIP encoder ([Zhai et al., 2023](#)) of SmolVLM ([Marafioti et al., 2025](#)) and inject cultural signal into the decoder via prompting and LoRA ([Hu et al., 2022](#)). A complementary line queries visual content before captioning: the CIC framework ([Yun and Kim, 2024](#)) extracts cultural elements via VQA and synthesizes a description with an LLM. Our cultural-VQA module is a direct descendant, but specializes the question set to the Americas and, crucially, repurposes its outputs as retrieval keys rather than as terminal captions. [Li et al. \(2026\)](#) (RAVENEA) show that *culture-aware retrieval* improves lightweight VLMs by $\geq 3.2\%$ on cultural VQA and 6.2% on cultural captioning, motivating our design in which cultural grounding steers—rather than runs parallel to—retrieval. Finally, category-stratified benchmarks (CulturalVQA, [Nayak et al., 2024](#); GIMMICK, [Schneider et al., 2025](#)) report that tangible artifacts are grounded more reliably than intangible practices, giving us a prior for RQ3 and, with cultural theory ([Yadav et al., 2025](#)), a basis for our human-evaluation rubric.

2.2 Cultural Taxonomies

[Hershcovich et al. \(2022\)](#) frame culture in NLP along linguistic form, values, common ground, and aboutness, arguing that serving a language means serving its cultural context—captioning in Guaraní or Bribri is thus a cultural grounding problem, not only translation. [Liu et al. \(2025\)](#) offer a finer taxonomy grounded in anthropology and survey where NLP resources are missing (non-Western, endangered, oral communities). We operationalize these descriptive frameworks as a small set of *visual question prompts* (ceremony, material culture, landscape, kinship) derived, following CIC ([Yun and Kim, 2024](#)), from ethnographic documentation of the target communities.

2.3 Retrieval, Prompting, and Evaluation

[Agrawal et al. \(2023\)](#) show that semantic similarity and diversity of in-context examples both matter for MT, motivating retrieval by cultural concept rather

than surface text. [Merx et al. \(2024\)](#) obtain strong low-resource gains from retrieval-augmented LLM prompting for Mambai but document a sharp in-domain vs. out-of-domain gap—directly relevant given that our indigenous corpora are drawn from formal/religious registers ([Mager et al., 2021](#)). Our retrieval uses multilingual Sentence-BERT embeddings ([Reimers and Gurevych, 2020](#)), which [Nie et al. \(2022\)](#) find outperform sparse BM25 for morphologically rich low-resource pairs. For evaluation we adopt ChrF++ ([Popović, 2017](#)), the official CICIL metric ([AmericasNLP 2026 Organizers, 2026](#)), which awards partial credit for morphological variants; [Kumar et al. \(2026\)](#) caution that character metrics can mask hallucination, source-copying, and repetition, which we corroborate below. Our reporting layout follows per-language templates such as SemEval-2025 Task 11 ([Muhammad et al., 2025](#)) and the AmericasNLP translation tasks ([Ebrahimi et al., 2023, 2024](#)), and our human-evaluation design follows the transparent, rubric-piloted protocol of THumB ([Kasai et al., 2022](#)).

3 Data and Exploratory Analysis

The CICIL pilot/dev data are extremely small: the dev split provides 50 images each for Guaraní, Bribri, Yucatec Maya, and Nahuatl, while Wixárika is supplied only as a 20-image *pilot*—and that pilot is the sole split carrying gold Spanish captions (used as our Stage 1 proxy). Stage 2 retrieval banks come from prior shared tasks and vary by orders of magnitude (Table 1): Guaraní is best resourced ($\sim 53k$ pairs, [Ebrahimi et al., 2023](#)), Bribri and Wixárika far smaller ($\sim 7.5k/\sim 9k$, [Mager et al., 2021](#)), and Maya/Nahuatl banks are still being assembled. Caption lengths differ substantially (Guaraní 21.6 words on average vs. Maya 11.3); because the targets differ in morphological typology, raw word counts are not comparable across languages, reinforcing our use of character-level ChrF++ and cautioning against cross-language score comparison.

4 System

4.1 Stage 1: Culturally-Aware Vision

Stage 1 turns an image into a culturally grounded Spanish description. The **cultural-VQA** module first asks category-specific questions (ceremony, material culture, landscape, kinship), then synthesizes the answers into a single concise Span-

Language	Dev	Cap. len	Bank pairs
Guaraní	50	21.6	53,183
Bribri	50	14.9	7,506
Yucatec Maya	50	11.3	TBD
Nahuatl	50	—	TBD
Wixárika	20 [†]	13.7	8,967

Table 1: Exploratory analysis. “Dev” = examples in the split; “Cap. len” = mean gold caption length (words); “Bank pairs” = Stage 2 retrieval corpus size reported by the source shared task. [†]Wixárika ships as a 20-image pilot (the only split with gold Spanish).

ish sentence and retains the per-category answers as structured annotations. Our primary model is SmolVLM-2B (Marafioti et al., 2025): we freeze its SigLIP encoder (Zhai et al., 2023) and adapt only the language decoder with LoRA (Hu et al., 2022) ($r=16$, $\alpha=32$, dropout 0.05), which is tractable on a single T4 within a \$50 budget; the frozen-encoder design is exactly why cultural signal must enter through prompting rather than encoder re-alignment. As the RQ1 control we run an off-the-shelf Qwen2.5-VL (Qwen Team, 2025) with no cultural prompting. For these preliminary numbers, both arms are produced with Qwen2.5-VL-7B served locally; the LoRA fine-tune is in progress.

4.2 Stage 2: Culturally-Indexed Retrieval and Translation

Stage 2 encodes the Spanish side of each parallel corpus with paraphrase-multilingual-MiniLM-L12-v2 (Reimers and Gurevych, 2020) into a FAISS inner-product index. The key design choice is that queries are built from Stage 1’s *cultural annotations* (for the generic arm, from the description itself), so retrieval targets cultural rather than surface similarity (Agrawal et al., 2023; Li et al., 2026). The top- k Spanish–indigenous pairs are supplied as few-shot demonstrations—most similar last (Agrawal et al., 2023)—to Gemini 2.5 Flash, which translates the Stage 1 Spanish into the target language. No LLM fine-tuning is performed. We score with sacreBLEU ChrF++ (word_order=2) matching the official CICIL scorer.

5 Preliminary Results

Official baseline (reference). Recomputed with the official scorer, the shared-task baseline reaches 20.82 ChrF++ (Guaraní), 17.77 (Wixárika), 11.53

Stage 1 configuration	ChrF++	len	conflicts
Generic baseline	21.0	35	2/20
Cultural-VQA, verbose	16.2	105	—
Cultural-VQA, concise	20.9	25	0/20

Table 2: Stage 1 intrinsic quality on the Wixárika pilot ($n=20$), Qwen2.5-VL-7B, temperature 0, ChrF++ vs. gold Spanish. “len” = mean length (tokens); “conflicts” = descriptions naming incompatible materials.

(Nahuatl), and 7.57 (Bribri); Maya has no MT baseline. These are the reference points in Table 3.

Stage 1 intrinsic quality. Because the authors do not read the target languages, we first evaluate Stage 1 on its Spanish output against gold Spanish on the Wixárika pilot ($n=20$), isolating vision-and-grounding from translation (Table 2). A first cultural-VQA implementation *underperformed* the generic baseline (16.2 vs. 21.0), which diagnostic analysis attributed to verbosity dilution—long, hedged syntheses (mean 105 tokens vs. ~ 35) penalized by character-F against short references. Constraining synthesis to one concise sentence (mean 25 tokens) closed the gap (20.9 vs. 21.0, $\Delta=-0.15$; parity) while removing internally inconsistent descriptions (0/20 vs. 2/20 naming incompatible materials). On this proxy, concise cultural-VQA matches the generic baseline without degrading it.

End-to-end (RQ1). Table 3 reports dev-set ChrF++ for the full pipeline in both arms at $k=5$. **Cultural-VQA does not yield a consistent end-to-end improvement.** Four of five languages fall within ± 0.6 ChrF++ of the generic control—within run-to-run noise for $n=50$ and therefore indistinguishable—and only Nahuatl shows a clear gain (+1.56); the mean Δ across languages is +0.20. Two sanity checks pass: our generic Guaraní (21.26) edges the official baseline (20.82) and generic Nahuatl (13.96) exceeds its baseline (11.53), so the pipeline is behaving sensibly rather than producing noise.

Interpretation. Three caveats bound these numbers. First, a genuine retrieval bank exists *only* for Wixárika (the 20-pair pilot); the other four languages were translated zero-shot because their parallel banks are not yet integrated, so the k -ablation is a single point ($k=5$) and the retrieval contribution is untested for four of five languages. Second, greedy decoding degenerated into repeti-

Language	Base	Gen.	Cult.	Δ
Guaraní	20.82	21.26	20.70	-0.56
Bribri	7.57	4.91	4.41	-0.50
Yucatec Maya	–	19.25	19.56	+0.30
Wixárika	17.77	8.97	9.19	+0.22
Nahuatl	11.53	13.96	15.52	+1.56
Mean Δ				+0.20

Table 3: End-to-end dev ChrF++ ($k=5$). “Base” = official baseline; “Gen.”/“Cult.” = generic vs. cultural-VQA arms; Δ = Cult. – Gen. Four of five deltas are within noise (± 0.6); only Nahuatl moves clearly.

tion loops on the lowest-resource languages (Bribri ~ 5 ChrF++ in both arms; Nahuatl’s *generic* arm frequently looped), which we bounded with an output-length cap—an honest RQ2 signal that a general model cannot translate these languages without a real bank. Third, and consequently, Nahuatl’s +1.56 is confounded: cultural-VQA there partly wins by avoiding a pathologically degenerate generic baseline rather than by demonstrably better cultural grounding. Disentangling genuine grounding from reduced degeneration is precisely what the planned human evaluation and category breakdown (RQ3) are designed to do.

6 Conclusion

We presented a two-stage, culturally-aware captioning pipeline and its first end-to-end numbers. On an intrinsic proxy, concise cultural-VQA matches a generic baseline; end-to-end, it is presently indistinguishable from the control on four of five languages, with one confounded gain. These results, together with an incomplete retrieval bank, define the controlled experiments we prioritize next: integrating the full parallel banks, running the LoRA fine-tune, and the category-stratified and human evaluations that address RQ2 and RQ3.

Limitations

Our targets are morphologically rich, low-resource languages; results are on small dev/pilot splits ($n \leq 50$), so ChrF++ differences below roughly one point are not meaningful and we report them as such. The cultural-VQA arm’s end-to-end contribution is confounded by (i) a retrieval bank present for only one language, (ii) reliance on a proprietary translator (Gemini 2.5 Flash) whose behavior may change, and (iii) greedy-decoding degeneration on the lowest-resource languages. The Stage 1 intrinsic evaluation exists only for Wixárika (the

sole split with gold Spanish), and the SmolVLM LoRA fine-tune is not yet complete, so the reported arms use an off-the-shelf VLM. Several teammate-supplied citations are transcribed from draft reference lists and marked for verification.

Ethics Statement

The CICIL data are released under CC BY-NC 4.0. To respect the NonCommercial clause we translate via Vertex AI, which does not use inputs to train models, rather than a consumer API tier that may; our use is non-commercial academic research and we attribute the dataset. This work aims to support, not supplant, indigenous-language communities; captions generated by our system are not authoritative and should be validated by native and heritage speakers before any community-facing use. We report negative and confounded results transparently to avoid overstating the utility of automated cultural captioning.

References

- Sweta Agrawal, Chunting Zhou, Mike Lewis, Luke Zettlemoyer, and Marjan Ghazvininejad. 2023. [In-context examples selection for machine translation](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8857–8873. Association for Computational Linguistics.
- AmericasNLP 2026 Organizers. 2026. The AmericasNLP 2026 shared task on cultural image captioning for indigenous languages (CICIL). https://americasnlp.org/2026_st.html. Shared task and dataset; see also <https://github.com/AmericasNLP/americasnlp2026>.
- Aashish Dhawan, Christopher Driggers-Ellis, Dzmitry Kasinets, Daisy Zhe Wang, and Christan Grant. 2026. [Retrieval-augmented long-context translation for cultural image captioning: Gators submission for AmericasNLP 2026 shared task](#). In *Proceedings of the Sixth Workshop on NLP for Indigenous Languages of the Americas (AmericasNLP)*.
- Abteen Ebrahimi, Ona de Gibert, Raúl Vázquez, Rolando Coto-Solano, Pavel Denisov, Robert Pugh, Manuel Mager, Arturo Oncevay, Luis Chiruzzo, Katharina von der Wense, and Shruti Rijhwani. 2024. [Findings of the AmericasNLP 2024 shared task on machine translation into indigenous languages](#). In *Proceedings of the 4th Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP)*, pages 236–246. Association for Computational Linguistics.
- Abteen Ebrahimi, Manuel Mager, Adam Wiemerslage, Pavel Denisov, Arturo Oncevay, Katharina von der Wense, et al. 2023. [Findings of the AmericasNLP](#)

- 2023 shared task on machine translation into indigenous languages. In *Proceedings of the Third Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP)*. Association for Computational Linguistics.
- Daniel Hershcovich, Stella Frank, Heather Lent, Miryam de Lhoneux, Mostafa Abdou, Stephanie Brandl, Emanuele Bugliarello, Laura Cabello Piqueras, Ilias Chalkidis, Alessandra Teresa Cignarella, et al. 2022. [Challenges and strategies in cross-cultural NLP](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations (ICLR)*.
- Yuchen Huang, Zhiyuan Fan, Zhitao He, Sandeep Polisetty, Wenyan Li, and Yi R. Fung. 2025. [CultureCLIP: Empowering CLIP with cultural awareness through synthetic images and contextualized captions](#). In *Conference on Language Modeling (COLM)*.
- Jungo Kasai, Keisuke Sakaguchi, Lavinia Dunagan, Jacob Morrison, Ronan Le Bras, Yejin Choi, and Noah A. Smith. 2022. [Transparent human evaluation for image captioning](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. Association for Computational Linguistics.
- Sanjeev Kumar, Preethi Jyothi, and Pushpak Bhat-tacharyya. 2026. [Evaluating extremely low-resource machine translation: A comparative study of chrF++ and BLEU metrics](#). *arXiv preprint arXiv:2602.17425*.
- Jiaang Li, Yifei Yuan, Wenyan Li, Mohammad Alian-nejadi, Daniel Hershcovich, Anders Søgaard, Ivan Vulić, Wenxuan Zhang, Paul Pu Liang, Yang Deng, and Serge Belongie. 2026. [RAVENE: A benchmark for multimodal retrieval-augmented visual culture understanding](#). In *International Conference on Learning Representations (ICLR)*.
- Chen Cecilia Liu, Iryna Gurevych, and Anna Korhonen. 2025. [Culturally aware and adapted NLP: A taxonomy and a survey of the state of the art](#). *Transactions of the Association for Computational Linguistics*, 13:652–689.
- Fangyu Liu, Emanuele Bugliarello, Edoardo Maria Ponti, Siva Reddy, Nigel Collier, and Desmond Elliott. 2021. [Visually grounded reasoning across languages and cultures](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics.
- Manuel Mager, Arturo Oncevay, Abteen Ebrahimi, John Ortega, Annette Rios, Angela Fan, Ximena Gutierrez-Vasques, Luis Chiruzzo, Gustavo Giménez-Lugo, Ricardo Ramos, et al. 2021. [Findings of the Americas-NLP 2021 shared task on open machine translation for indigenous languages of the americas](#). In *Proceedings of the First Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP)*, pages 202–217. Association for Computational Linguistics.
- Andrés Marafioti et al. 2025. [SmolVLM: Redefining small and efficient multimodal models](#). *arXiv preprint arXiv:2504.05299*.
- Raphaël Merx, Aso Mahmudi, Katrina Langford, Leo Alberto de Araujo, and Ekaterina Vylomova. 2024. [Low-resource machine translation through retrieval-augmented LLM prompting: A study on the mambai language](#). In *Proceedings of the 2nd Workshop on the Interplay of Endangered Languages and AI (EURALI) at LREC-COLING 2024*. ELRA and ICCL.
- Shamsuddeen Hassan Muhammad, Nedjma Ousidhoum, Idris Abdulmumin, et al. 2025. [SemEval-2025 task 11: Bridging the gap in text-based emotion detection](#). In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Association for Computational Linguistics.
- Shravan Nayak, Kanishk Jain, Rabiul Awal, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stanczak, and Aishwarya Agrawal. 2024. [Benchmarking vision language models for cultural understanding](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5769–5790. Association for Computational Linguistics.
- Ercong Nie, Sheng Liang, Helmut Schmid, and Hinrich Schütze. 2022. [Cross-lingual retrieval augmented prompt for low-resource languages](#). *arXiv preprint arXiv:2212.09651*.
- Maja Popović. 2017. [chrF++: words helping character n-grams](#). In *Proceedings of the Second Conference on Machine Translation (WMT)*, pages 612–618. Association for Computational Linguistics.
- Qwen Team. 2025. [Qwen2.5-VL technical report](#). *arXiv preprint arXiv:2502.13923*.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *PMLR*, pages 8748–8763.
- Nils Reimers and Iryna Gurevych. 2020. [Making monolingual sentence embeddings multilingual using knowledge distillation](#). In *Proceedings of the*

2020 *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4512–4525. Association for Computational Linguistics.

David Romero et al. 2024. [CVQA: Culturally-diverse multilingual visual question answering benchmark](#). In *Advances in Neural Information Processing Systems 37 (NeurIPS): Datasets and Benchmarks Track*.

Florian Schneider, Carolin Holtermann, Chris Biemann, and Anne Lauscher. 2025. [GIMMICK: Globally inclusive multimodal multitask cultural knowledge benchmarking](#). In *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics.

Srishti Yadav, Lauren Tilton, Maria Antoniak, Taylor Arnold, Jiaang Li, et al. 2025. [Cultural evaluations of vision-language models have a lot to learn from cultural theory](#). *arXiv preprint arXiv:2505.22793*.

Youngsik Yun and Jihie Kim. 2024. [CIC: A framework for culturally-aware image captioning](#). In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)*.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. [Sigmoid loss for language image pre-training](#). In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.